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SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCE

CHENNAI - 602105

AI-Based Genetic Disorder Risk Prediction and Classification

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ABSTRACT

- ✓ GeneGuard is an **AI-based system for predicting the risk of genetic disorders**.
- ✓ It analyzes selected genetic and patient-related features.
- ✓ Machine Learning is used to classify the user into different **risk levels**.
- ✓ Explainable AI helps users understand **why a particular risk prediction was made**.
- ✓ The system provides **counseling-oriented recommendations** based on the predicted risk.
- ✓ Genetic and personal information is handled with **privacy and security considerations**.
- ✓ GeneGuard is a **decision-support system and not a medical diagnosis tool**.

INTRODUCTION

- ✓ **Genetic disorders** are diseases caused by changes or abnormalities in genes or chromosomes.
- ✓ **Early risk prediction** can help identify individuals who may require further medical evaluation.
- ✓ **Artificial Intelligence (AI)** can analyze healthcare and genetic data to identify patterns related to disease risk.
- ✓ **Explainable AI (XAI)** helps users understand the important factors behind an AI-generated prediction.
- ✓ **GeneGuard** combines AI-based risk prediction with **explainable results and counseling support**.
- ✓ The system also focuses on **privacy and security** because genetic and personal data are highly sensitive.

OBJECTIVE

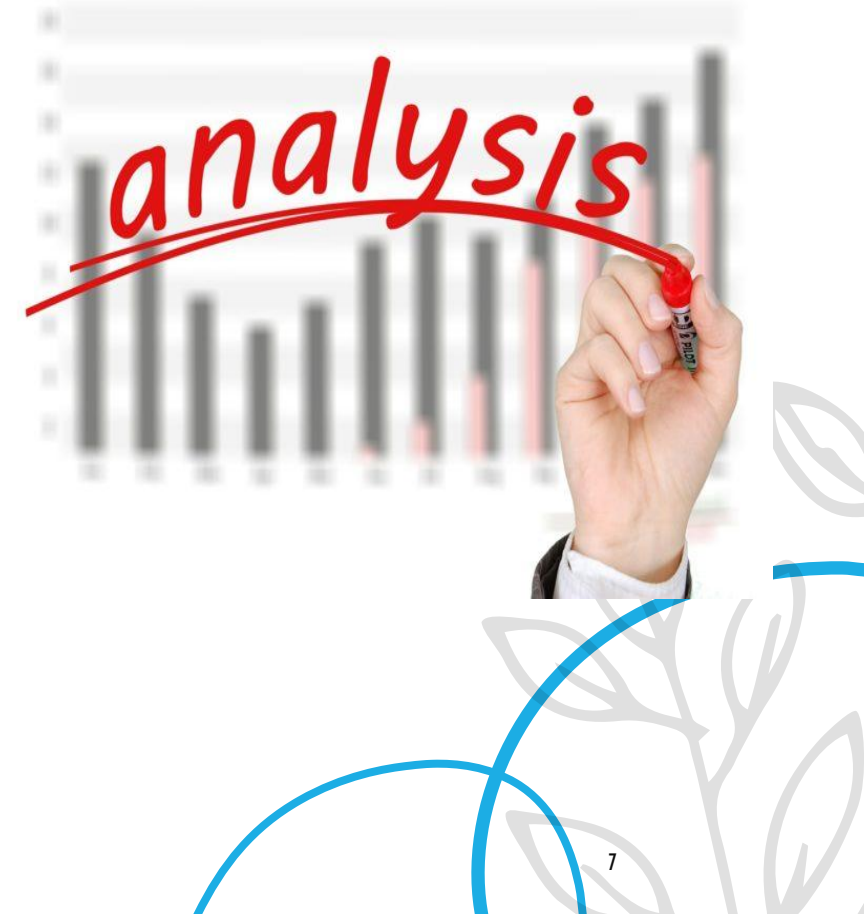
- ✓ To develop an **AI-based genetic disorder risk prediction system**.
- ✓ To classify users into suitable **risk categories**.
- ✓ To provide understandable explanations for AI predictions.
- ✓ To provide counseling-oriented recommendations.
- ✓ To protect sensitive genetic and personal information.
- ✓ To support healthcare professionals in decision-making.

LITERATURE SURVEY

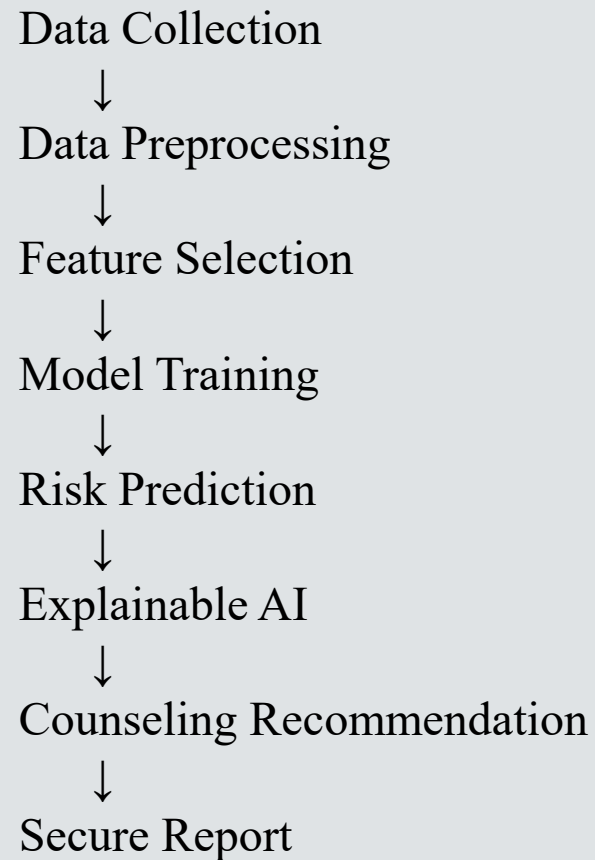
Author(s)	Title	Year	Methodology	Key Findings
V. Chen et al.	Interpretable Machine Learning in Computational Biology	2024	Explainable ML	Improves interpretability of biological data.
A. S. Alharthi et al.	Explainable AI in Human Health Monitoring	2024	XAI & Healthcare	Improves transparency in health prediction.
Q. Sun et al.	Explainable AI for Medical Applications	2024	XAI & Medical AI	Supports trustworthy medical predictions.
R. Alkhanbouli et al.	Explainable AI in Disease Prediction	2025	XAI & ML	SHAP and LIME improve prediction interpretation.
C. Wang et al.	Deep Learning and Explainable AI for Genetic Insights	2025	Deep Learning & XAI	Provides interpretable insights from genetic data.

PROBLEM STATEMENT :

- ✓ Genetic disorder risk assessment can involve complex information.
- ✓ Users may not understand how risk predictions are generated.
- ✓ Black-box AI models can reduce trust.
- ✓ Sensitive genetic information requires strong privacy protection.
- ✓ Existing systems may not provide personalized counseling support.
- ✓ Therefore, an explainable and secure AI-based risk prediction system is needed.



METHODOLOGY



System Architecture

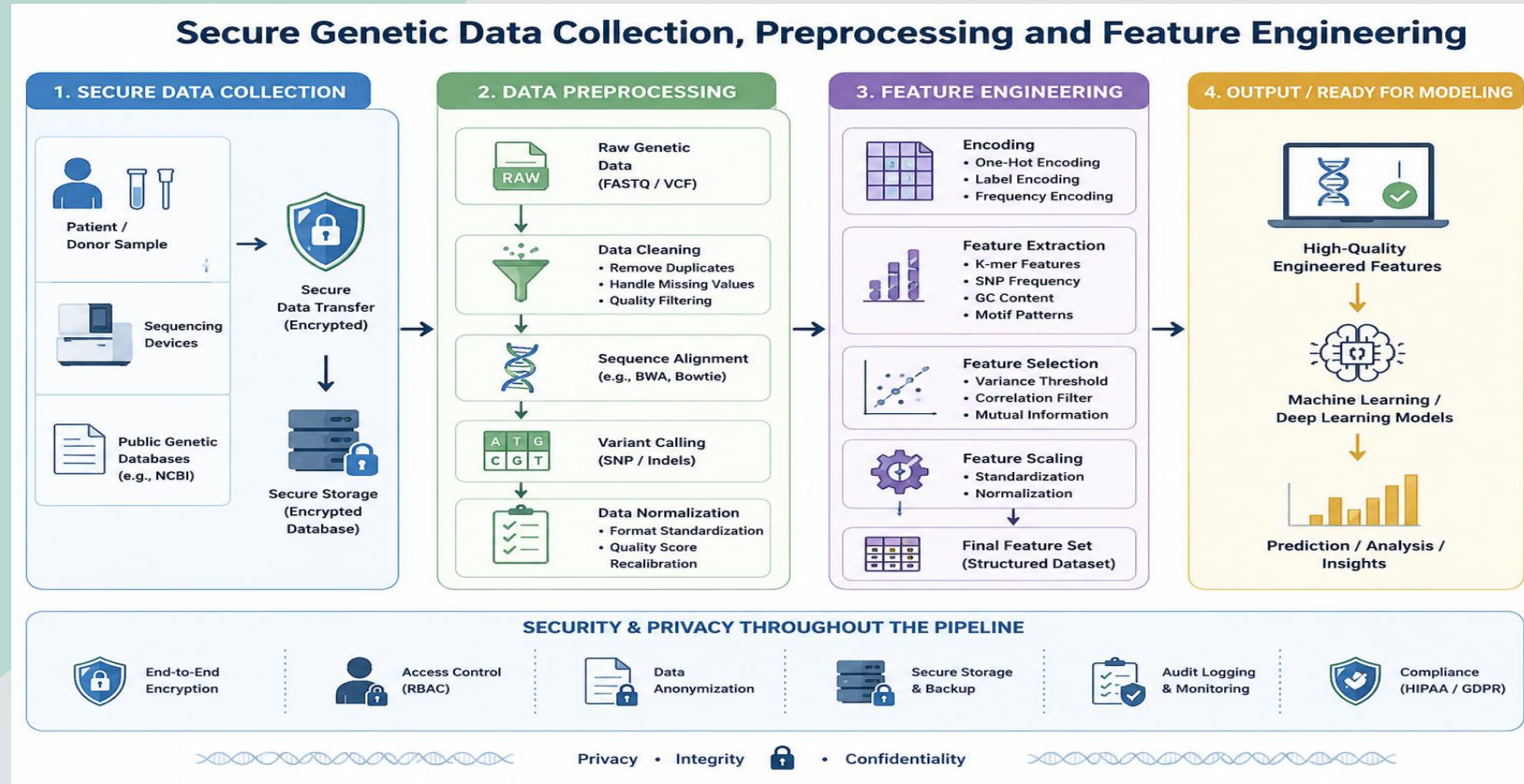
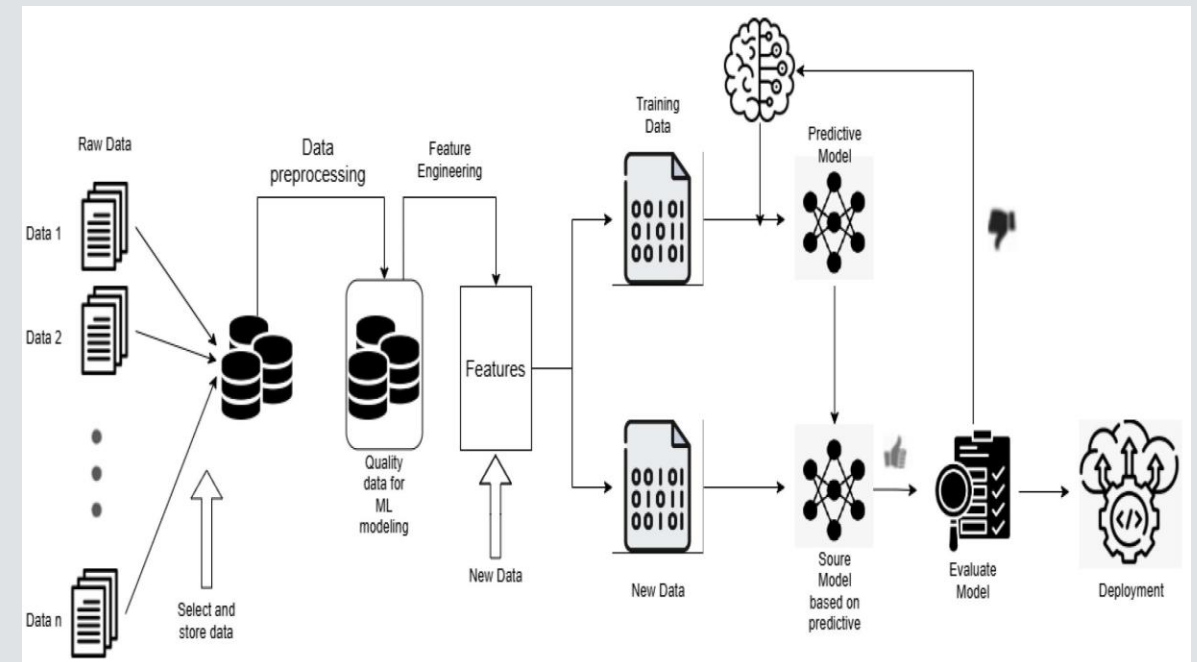


Figure 1 : Secure Genetic Data Collection, Preprocessing and Feature Engineering

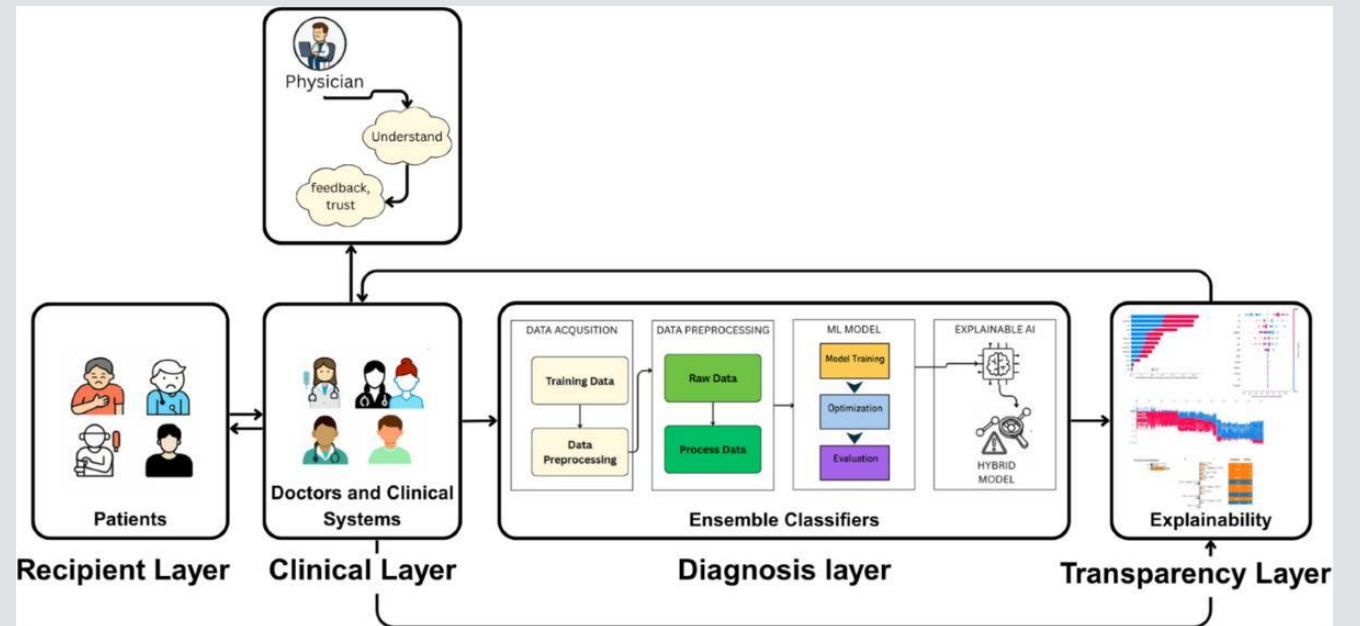
Module 1: Secure Patient Data Module

- ✓ User registration/login
- ✓ Secure data input
- ✓ Patient information management
- ✓ Authentication
- ✓ Data protection



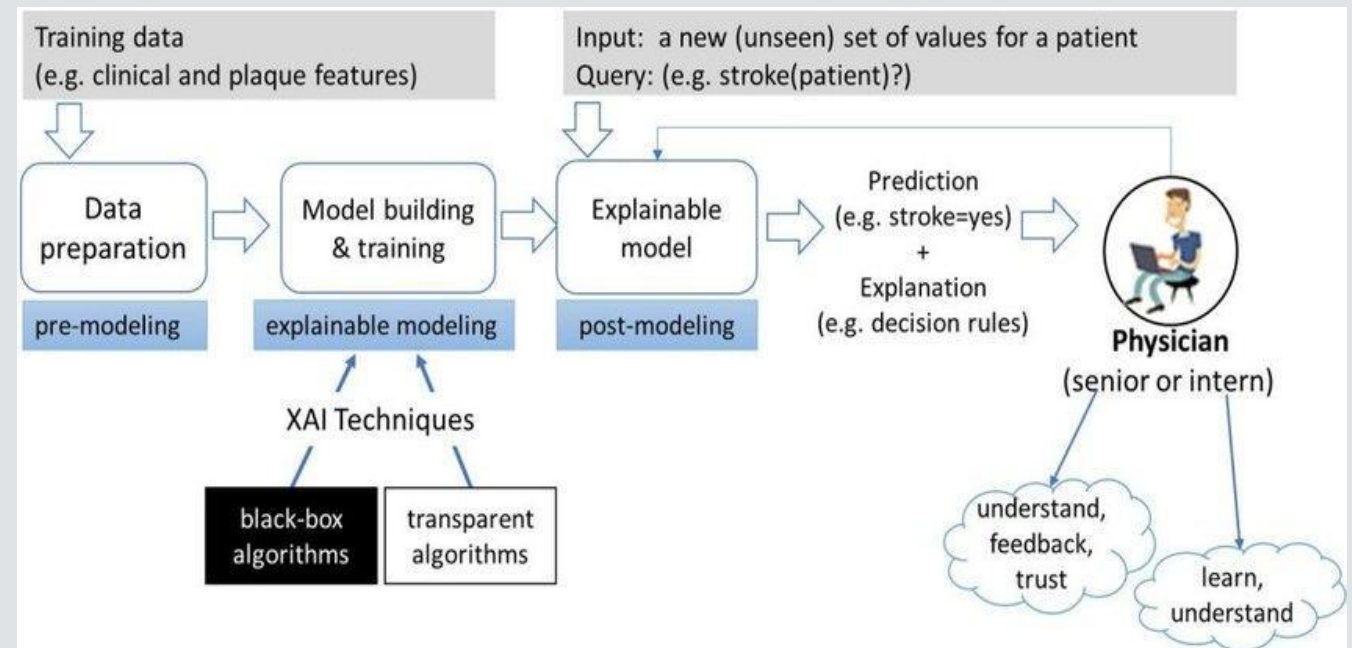
Module 2 – Genetic Risk Prediction Module

- ✓ Data preprocessing
- ✓ Feature selection
- ✓ ML model
- ✓ Risk classification
- ✓ Prediction generation



Module 3 – Explainable AI Module

- ✓ Analyze model prediction
- ✓ Identify important features
- ✓ Generate explanations
- ✓ SHAP/LIME-based interpretation
- ✓ Display understandable results



FUTURE ENHANCEMENT

Component	Technology
Programming	Python
Frontend	HTML, CSS, JavaScript / React
Backend	Flask / FastAPI
Machine Learning	Scikit-learn
Explainable AI	SHAP / LIME
Database	MySQL / SQLite
Security	Authentication + Encryption



CONCLUSION

- ✓ GeneGuard provides AI-assisted genetic disorder risk prediction.
- ✓ It converts complex data into understandable risk information.
- ✓ Explainable AI improves transparency and user trust.
- ✓ Counseling support helps users understand the next appropriate steps.
- ✓ Security mechanisms help protect sensitive information.
- ✓ The system supports professionals rather than replacing medical diagnosis.

REFERENCES :

- ✓ World Health Organization – Genetic Diseases and Genomics resources.
- ✓ Scikit-learn Documentation – Machine Learning.
- ✓ SHAP Documentation – Explainable AI.
- ✓ LIME – Local Interpretable Model-Agnostic Explanations.
- ✓ Research papers on Explainable AI in healthcare.
- ✓ Research papers on machine learning for genetic disorder prediction.

